

NT219- Cryptography

Introduction to Cryptology

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Warm up- Introduction to cryptography

1. Cipher Algorithms: Input vs Output?

<https://www.devglan.com/online-tools/aes-encryption-decryption>

2. Hash and HMAC: Input vs Output?

<https://emn178.github.io/online-tools/sha256.html>

3. Secure Network communication: https?

Debug: <https://www.charlesproxy.com/download/latest-release/>

4. Digital signature: Software distribution? Why?

Check some program on your PC

5. Hacking:

<https://hashkiller.io/listmanager>

- Introduction to cryptography courses

 - Why cryptology?

 - Syllabus;

 - Guide for reading textbooks and library;

 - Policy and course assessment;

 - Projects;

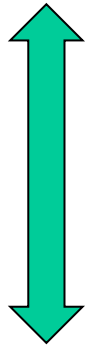
 - Labs and guides

Why cryptology?

Cryptology = Cryptography + Cryptanalysis

Information security

Reveal secrets, alter assets?



“Secret writing/
to protect digital assets”



Try to analysis the codes

Motivations

❖ Network devices:

- End devices
- Intermediate devices

❖ Communication links

- fiber, copper, radio
- transmission rate:
bandwidth

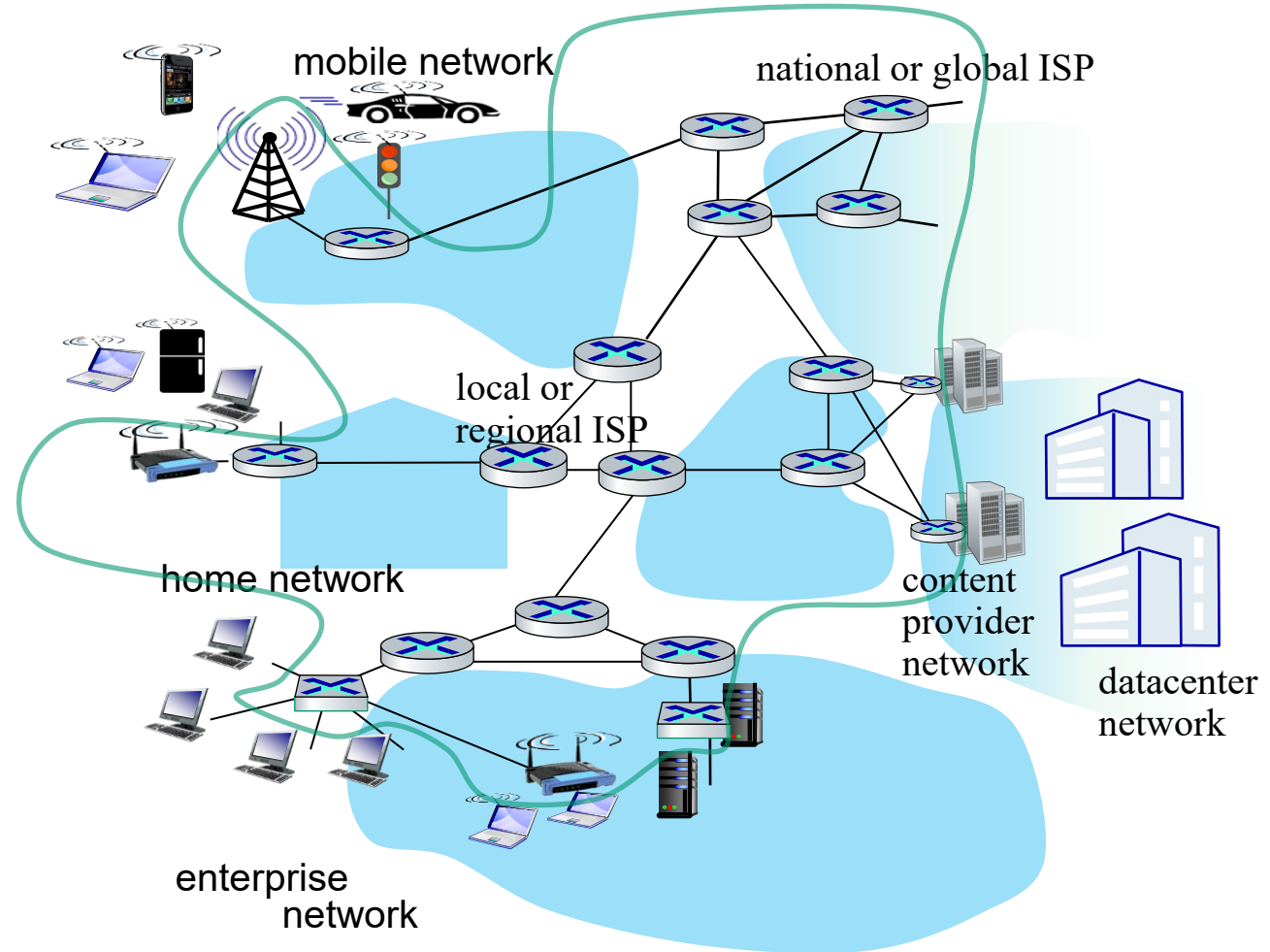
❖ Network protocols

- How to **secure** data?

Send? Receiver?

Storage?

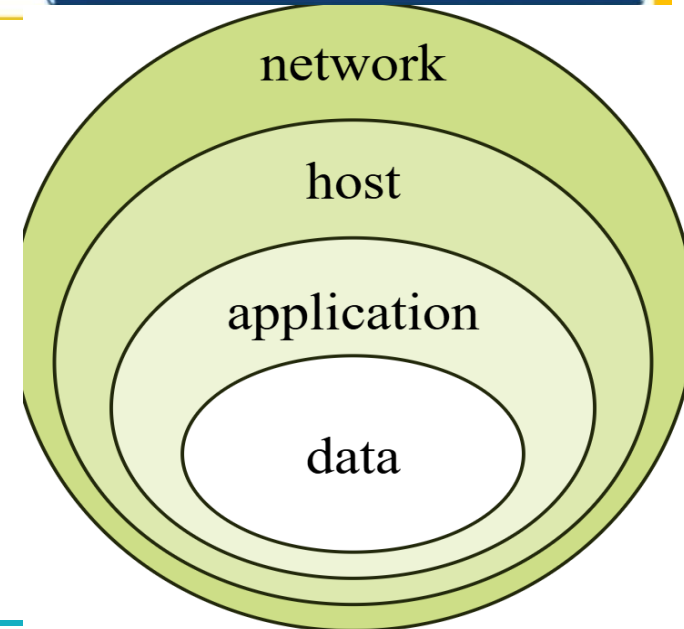
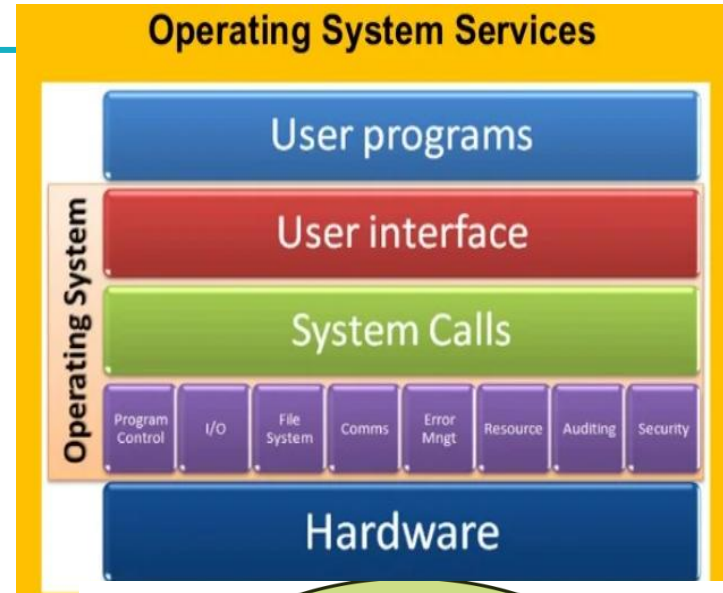
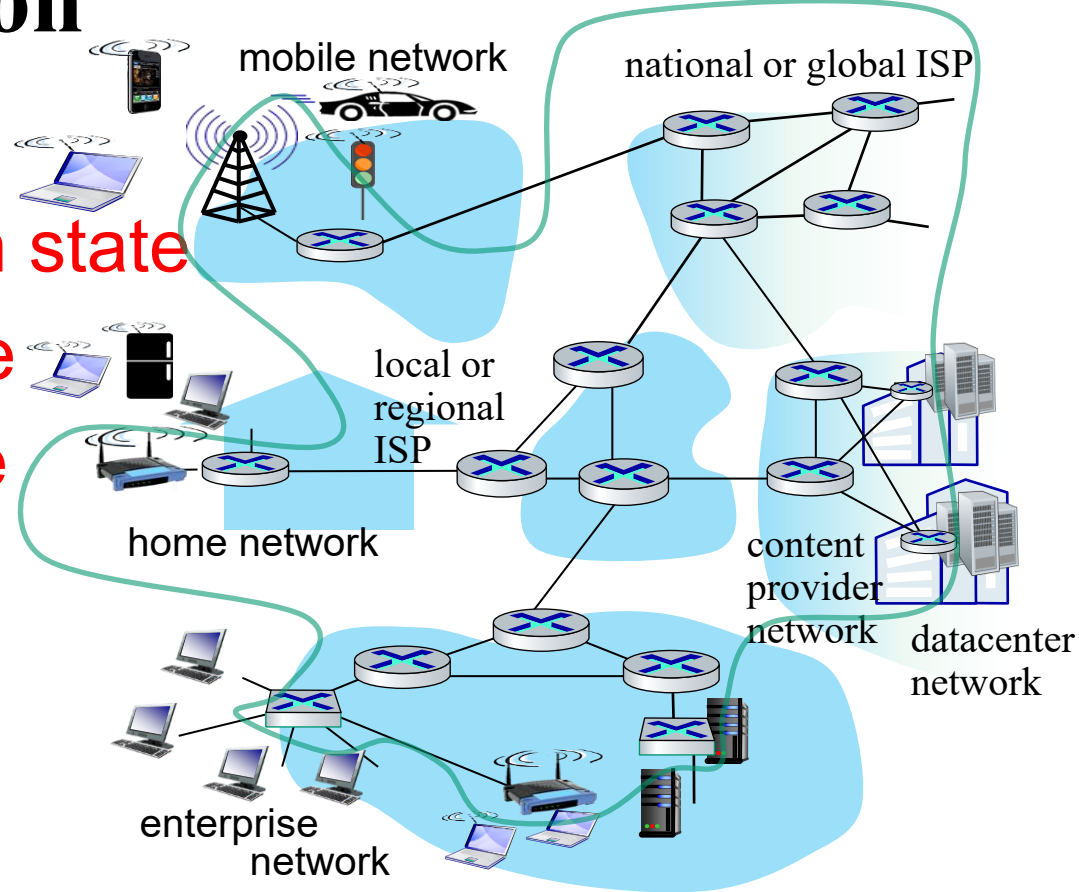
Process?



Data Protection

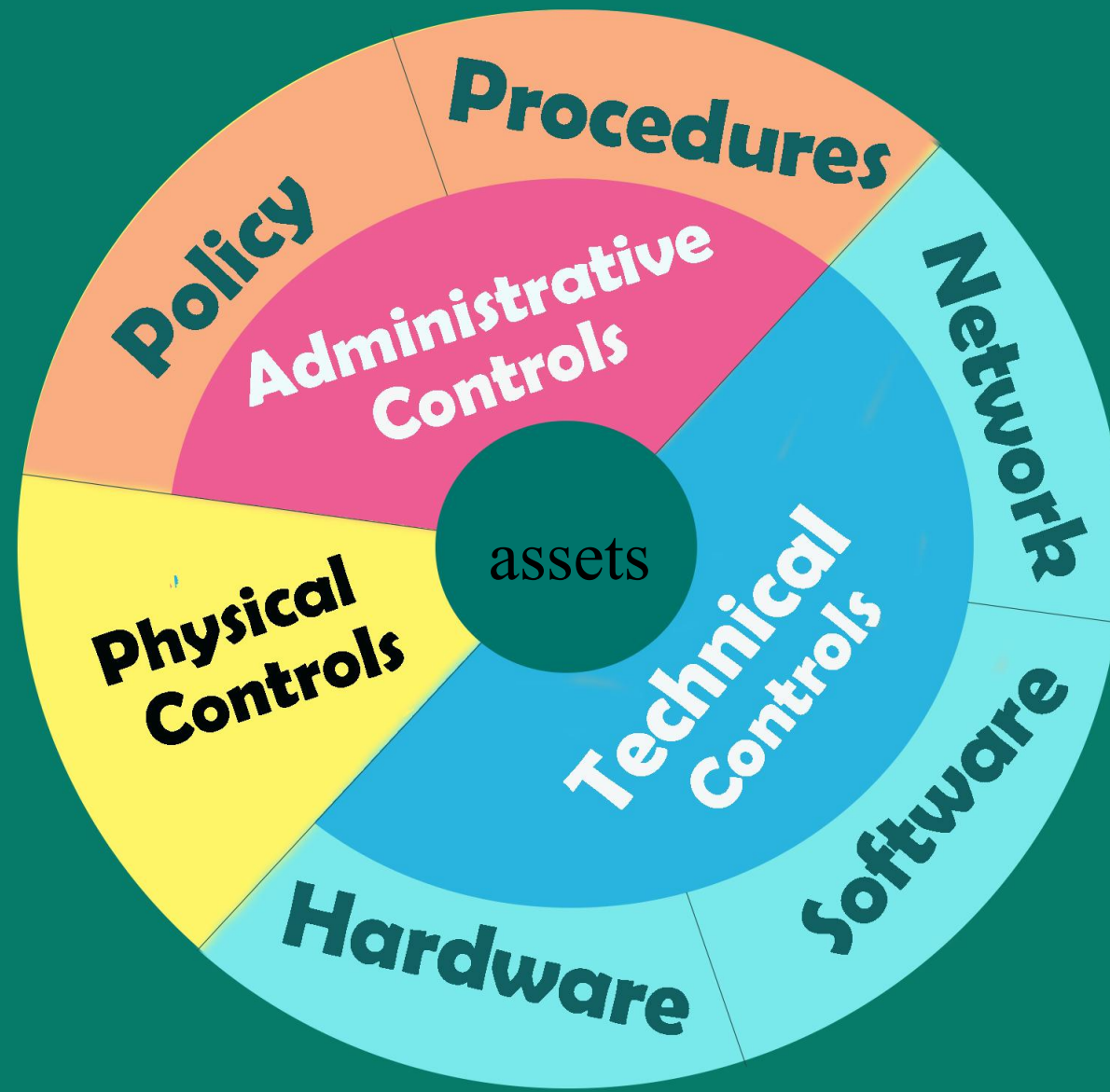
Data

- Transmission state
- Process state
- Storage state



Host vs Network?

Defense in Depth:



Technical solutions

Defense in depth:

Host or network?

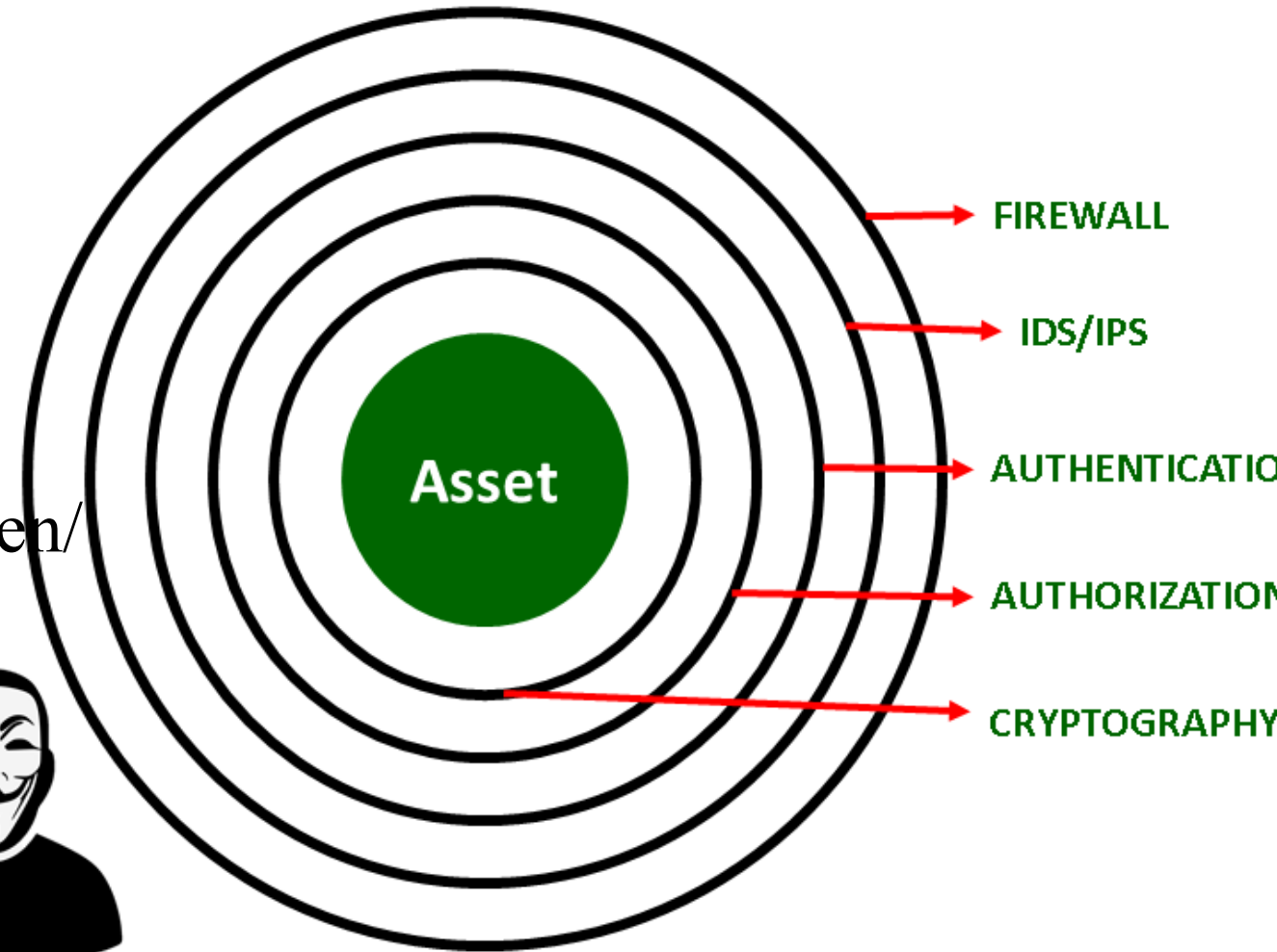
👉 <https://owasp.org/www-project-top-ten/>

- Internal threats
- External threats
- Partners



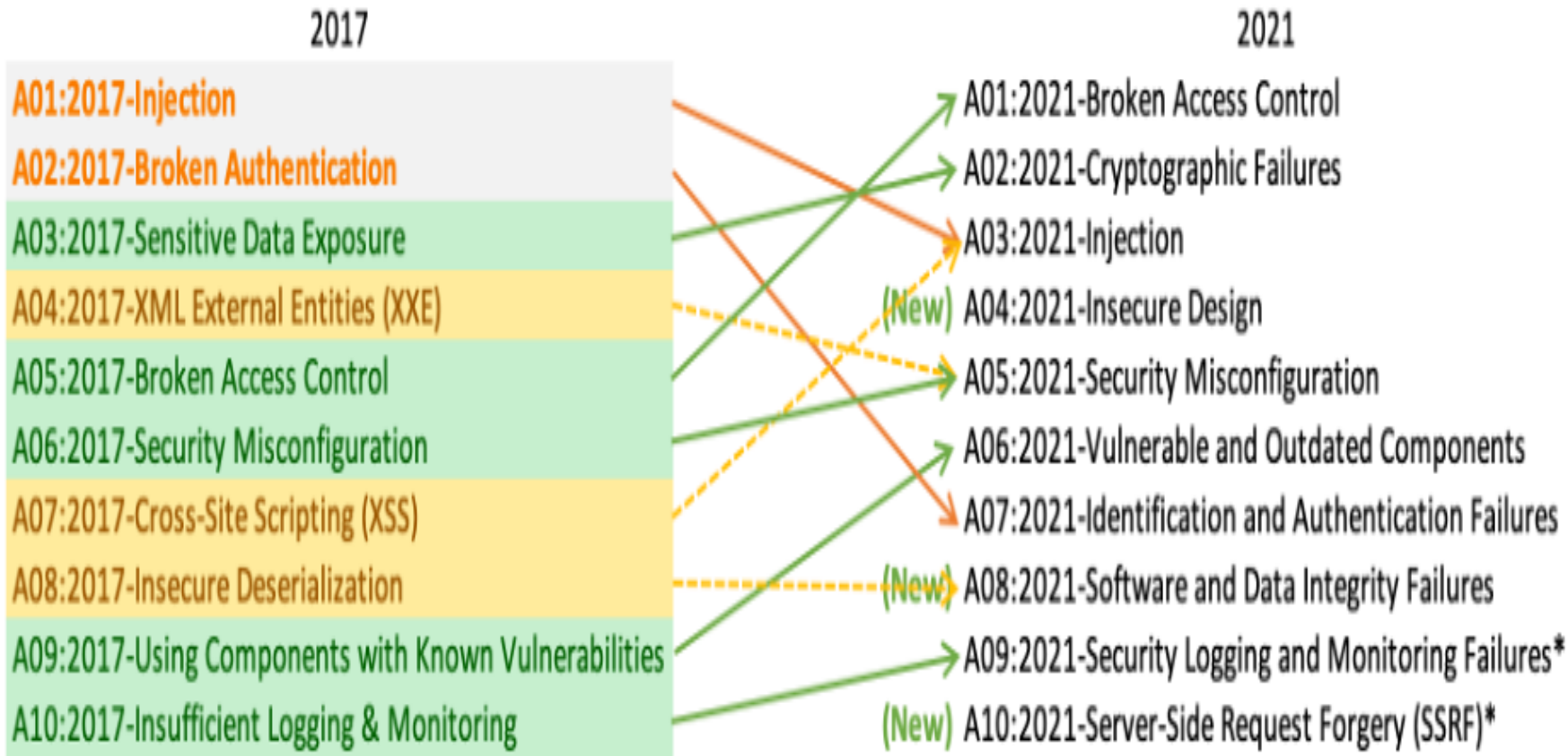
HACKER
adversary

ONION MODEL



Example solutions

👉 <https://owasp.org/www-project-top-ten/>



* From the Survey

Adversary

- Internal threats
- External threats
- Partners

What is cryptograph?

- Cryptology= Cryptography + Cryptanalysis

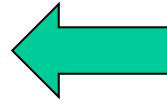
Goals

- Confidentiality
- Privacy

What?

Cipher systems

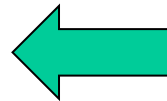
- Symmetric (AES)
- Asymmetric (RSA, ECC, CRYSTALS-KYBER)



- Integrity
- **Authentication**
- Non-repudiation (Accountability)

Hash functions

Message authentication code (MAC)
Digital signature (digital certificate)



- Availability

Motivations

Protect digital assets?

- Data and software
 - What is data?
 - Any object that can be processed or executed by a computer
 - Three states of data
 - Transmission state
 - Storage state
 - Processing state
 - Data vs. information security
- Hardware and devices
- Reputation

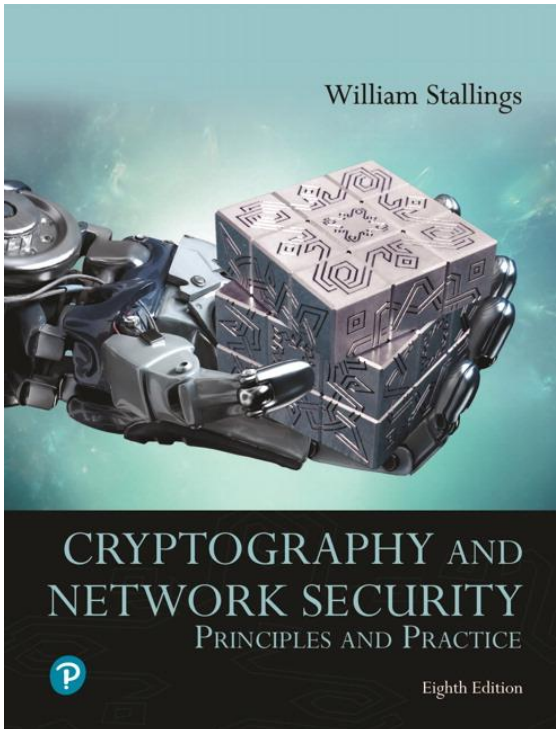
Motivations

Security goals?

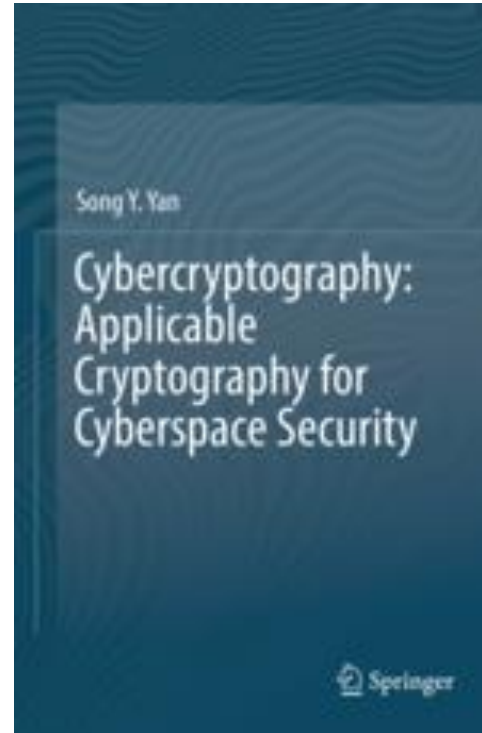
- **Confidentiality**
 - including data in transmission and storage states
- **Data integrity**
 - including data in transmission and storage state
- **Non-repudiation**
- **Authentication**
 - All nodes in network, users, ...
- **Privacy**
- **Availability**

Textbooks and References

■ Text books

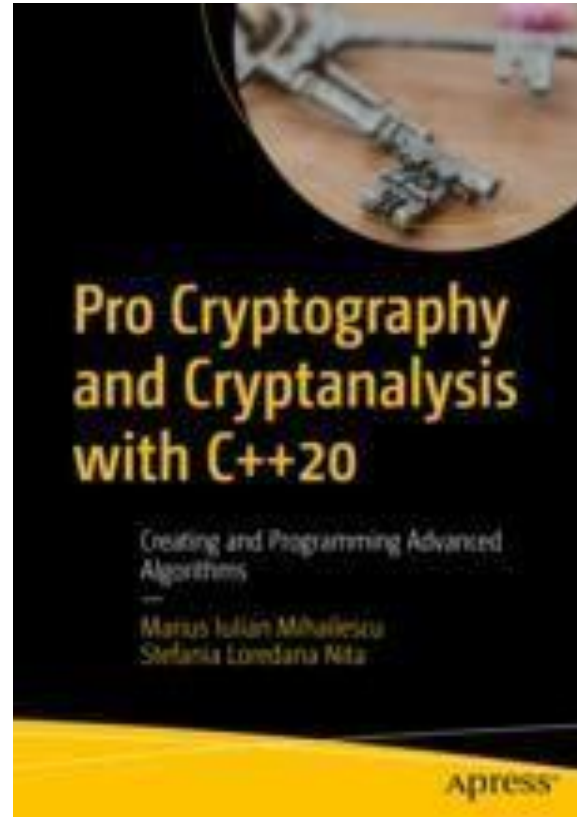


[1] Stallings, W. (2019). *Cryptography and network security : principles and practice (8th)*: Pearson Education



[2] Yan, S. Y.(2019). *Cybercryptography: Applicable Cryptography for Cyberspace Security*: Springer.

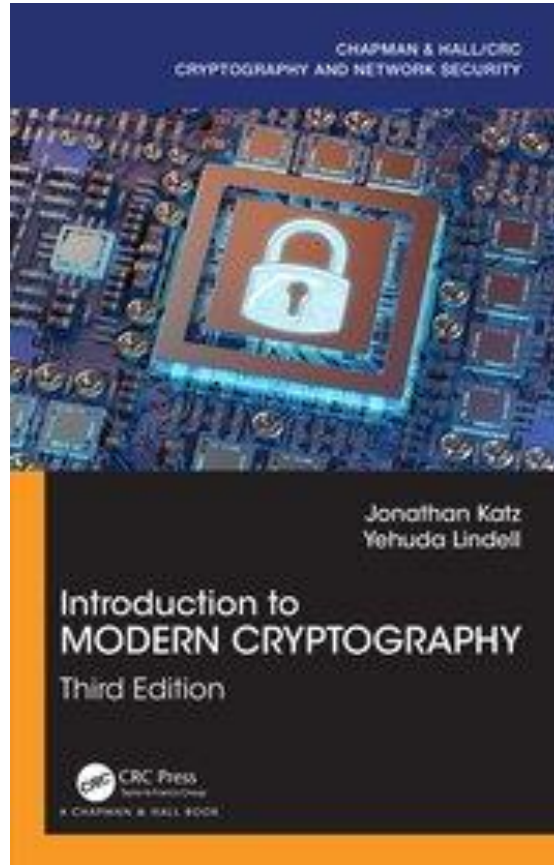
- Reference books for labs



[3] Mihailescu, M. I., & Nita, S. L. (2021). *Pro cryptography and cryptanalysis with C++20: creating and programming advanced algorithms*: Apress.

Textbooks and References

- Other references



[4] Katz, J., & Lindell, Y. (2020). *Introduction to modern cryptography* (3rd): CRC press.

Week 1. Introduction to the course

Week 2. Introduction to cryptography

- Cryptography overview;
- Some terminologies;
- Classical cryptography

Week 3,4. Symmetric cryptography

- Overview;
- Stream cipher (chaotic maps)
- Block cipher (DES, AES and mode of operations)
- Introduction to some other modern cipher systems

Week 5. Project reports (first time)

Week 6,7,8. Asymmetric cryptography

- Overview;
- Cryptosystems are based on factoring problems;
- Cryptosystems are based on discrete logarithm problems;
- Elliptic curve cryptosystems

Week 9,10. Hash function and data authentication

- Hash functions;
- HMAC;
- Data integrity;

Week 11. Digital signature

- Digital signature
- Application

Week 12,13. Applied cryptography

- Authentication (Kerberos);
- Network security (IPSec; SSL/TLS; SSH);
- Database security;

Week 14. Post-quantum Cryptography/ Blockchain network;

Week 15. Final project report;

❖ Textbooks

- [1] Stallings, W. (2019). *Cryptography and network security : principles and practice (8th)*: Pearson Education.
- [2] Yan, S. Y.(2019). *Cybercryptography: Applicable Cryptography for Cyberspace Security*: Springer.

❖ Labs

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❖ References

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Guide for reading textbooks and libraries

1. CrypTool 2: <https://www.cryptool.org/en/ct2/>
2. Crypto++ Library 8.9: <https://www.cryptopp.com/>
3. Openssl 3.1 <https://github.com/openssl/openssl>
3. C++ Programing Language: Visual Studio Code & MSYS2
<https://www.msys2.org/>;
<https://code.visualstudio.com/>
4. Windows and Linux OS:
Windows 10, 11; Ubuntu 24 or Kali Linux
<https://www.kali.org/>;
<https://releases.ubuntu.com/24.04/>

Policy and course assessment

1. **Attendants:** $\geq 80\%$;
2. **Assignments:** Must complete (Bonus points);
3. **Labs:** Must complete (home and classroom);
4. **CTF:** Bonus points
5. **Assessments:**
 - + **Process assessment (Project) 30%:**
15 (Final present) + 15 (final report);
 - + **Lab assessment: 20%;**
 - + **Final assessment: 50%**
Closed-book wiring exam or oral exam

Projects

1. **Project titles:** Chosen your-self
2. **References and literature survey:**
 - Management papers (Endnote)
 - Search and download papers
4. **Some other guides**
 - **Determine Topic**
 - **Problem analysis**
 - **Solution architecture**
 - **Implementation and testing**
 - **Application deployment**

6. References

- IEEE**
- ISI/Scopus**
- webofscience**
- Scholar**
- sci-hub.ru**

Lab guides

- Dual boot**
- Setup for windows**
- Setup for Linux**

Class communication

MS Team classroom

Class code: check your email (studentID@ms.uit.edu.vn)

My email: tunn@uit.edu.vn

Room: E8.6 (Thursday)